

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Due Date	02.01.2026		

Experiment 1: Working with Python packages – Numpy, Scipy, Scikit-Learn, Matplotlib

Aim:

To investigate and utilize Python libraries such as NumPy, Scikit-learn, and Matplotlib on public repository datasets to determine machine learning objectives, feature selection strategies, and appropriate predictive algorithms. **Libraries used:**

- NumPy (imported as `np`)
- Pandas (imported as `pd`)
- Matplotlib.pyplot (imported as `plt`)
- Seaborn (imported as `sns`)
- OpenCV (`cv2`)
- StandardScaler (from `sklearn.preprocessing`)
- Math (Standard Library)

Mathematical and Theoretical description of the objectives performed:

- **Purpose:** Conduct Exploratory Data Analysis (EDA) to investigate data characteristics, analyze distribution patterns, detect outliers, assess class balance, and distinguish dependencies between features.
- **Descriptive Statistics**
 - Arithmetic mean calculated via: $\mu = \frac{1}{n} \sum x_i$
 - Sample variance used to quantify data spread: $s^2 = \frac{1}{n-1} \sum (x_i - \mu)^2$
 - Quantile statistics (Median, $Q1$, $Q3$) employed to evaluate dispersion and generate box-plots.
- **Distribution Analysis Using Histograms**
 - Approximation of empirical distributions through frequency binning to illustrate data shape and central tendencies.

- Application of Kernel Density Estimation (KDE) to provide a continuous estimation of the probability density:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

where K represents the kernel function and h denotes the smoothing bandwidth.

- **Boxplot Visualization**

- Representation of the median and quartiles ($Q1$, $Q3$), defining the interquartile range as $IQR = Q3 - Q1$.
- Whiskers delineate data within the boundary $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$; values residing outside this range are identified as statistical outliers.

- **Feature Relationship Plots**

- Scatter plot matrices utilized for detecting trends, class separability, and non-linear correlations.
- Diagonal plots show individual feature distributions using histograms or KDEs.

- **Correlation Visualization**

- Quantification of linear relationships between variables using the Pearson correlation coefficient:

$$r_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{cov}(X, Y) = \frac{1}{n-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$$

- Coefficient values range from $[-1, 1]$, indicating both the strength and orientation of linear association.

- **Categorical Data Visualization**

- Frequency-based bar charts used to analyze category counts and investigate potential class imbalances.

- **Image Data Exploration**

- Analysis of image resolutions (height and width) via histograms or KDEs to ensure consistency for model input.
- Display of sample images to qualitatively evaluate visual data properties.

- **Handling Missing Data**

- Feature-wise audit of null values to determine appropriate mitigation strategies, such as data imputation or pruning.

- **Analytical Objectives**

- Identification of skewed data, multimodality, anomalous entries, collinear features, and class skews to facilitate effective preprocessing and model optimization.

Results and Discussions:

0.1 Loan Amount Prediction

	Customer ID	Name	Gender	Age	Income (USD)	Income Stability	Profession	Type of Employment	Location	Loan Amount Request (USD)	...	Credit Score	No. of Defaults	Has Active Credit Card	Property ID
0	C-36995	Frederica Shealy	F	56	1933.05	Low	Working	Sales staff	Semi-Urban	72809.58	...	809.44	0	NaN	746
1	C-33999	America Calderone	M	32	4952.91	Low	Working	NaN	Semi-Urban	46837.47	...	780.40	0	Unpossessed	606
2	C-3770	Rosetta Verne	F	65	988.19	High	Pensioner	NaN	Semi-Urban	45593.04	...	833.15	0	Unpossessed	546
3	C-26480	Zoe Chitty	F	65	NaN	High	Pensioner	NaN	Rural	80057.92	...	832.70	1	Unpossessed	896
4	C-23459	Afton Venema	F	31	2614.77	Low	Working	High skill tech staff	Semi-Urban	113858.89	...	745.55	1	Active	716
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
29995	C-43723	Angelyn Clevenger	M	38	4969.41	Low	Commercial associate	Managers	Urban	76657.90	...	869.61	0	Unpossessed	566
29996	C-32511	Silas Slaugh	M	20	1606.88	Low	Working	Laborers	Semi-Urban	66595.14	...	729.41	0	Inactive	176
29997	C-5192	Carmelo Lone	F	49	NaN	Low	Working	Sales staff	Urban	81410.08	...	NaN	0	Active	956
29998	C-12172	Carolann Osby	M	38	2417.71	Low	Working	Security staff	Semi-Urban	142524.10	...	677.27	1	Unpossessed	376
29999	C-33003	Bridget Garibaldi	F	63	3068.24	High	Pensioner	NaN	Rural	156290.54	...	815.44	0	Active	346

Figure 1: Dataset Columns

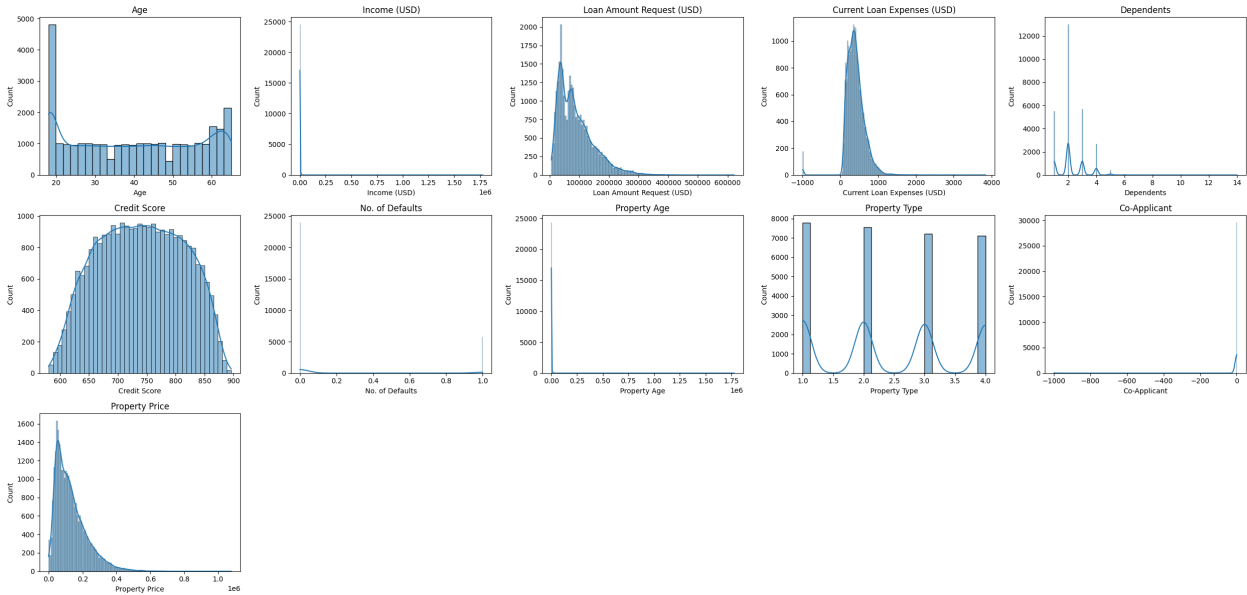


Figure 2: Loan Amount Distribution (Histogram Plot)

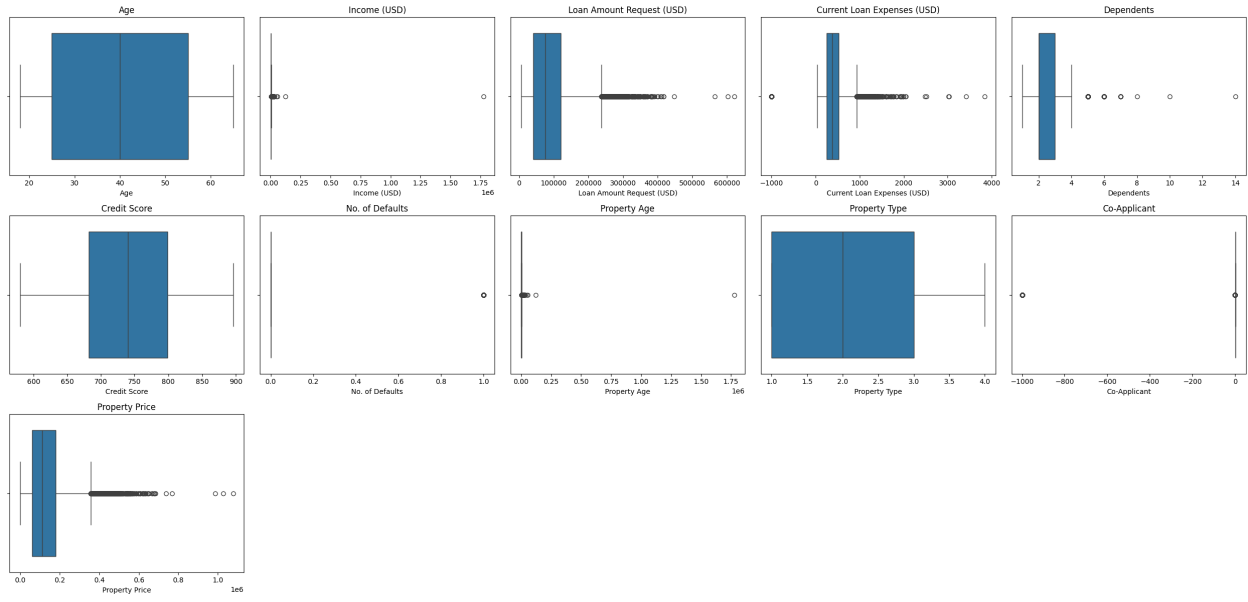


Figure 3: Boxplot Distribution

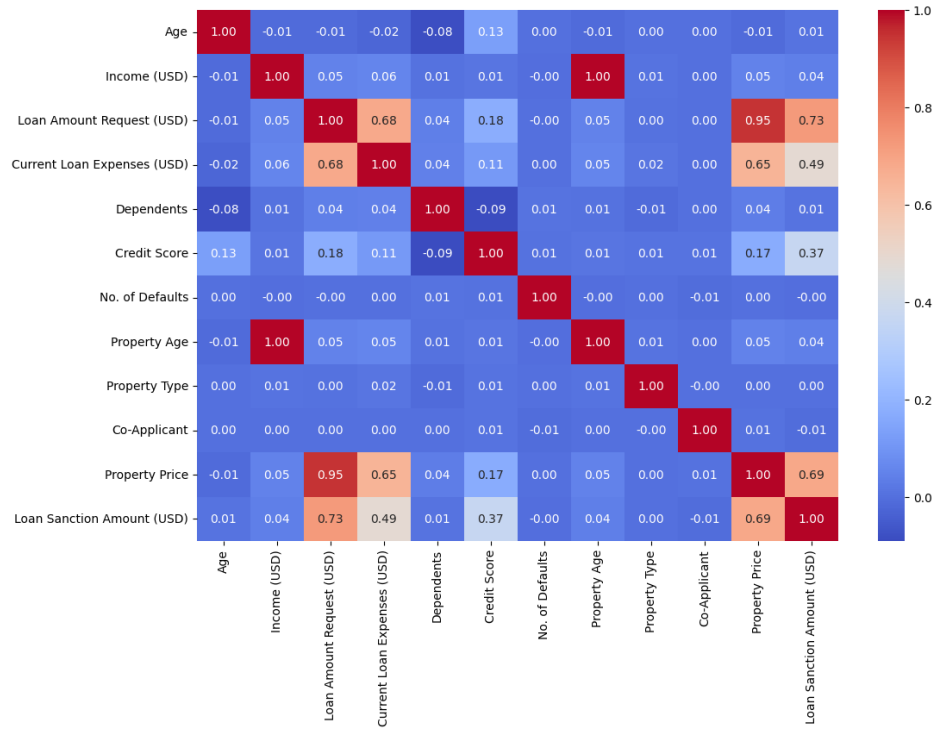


Figure 4: Correlation Matrix

0.2 Handwritten Character Recognition (MNIST)

0 1 2 3 4 5 6 7 8 9 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z a b c d e f g h i

Figure 5: Sample Images

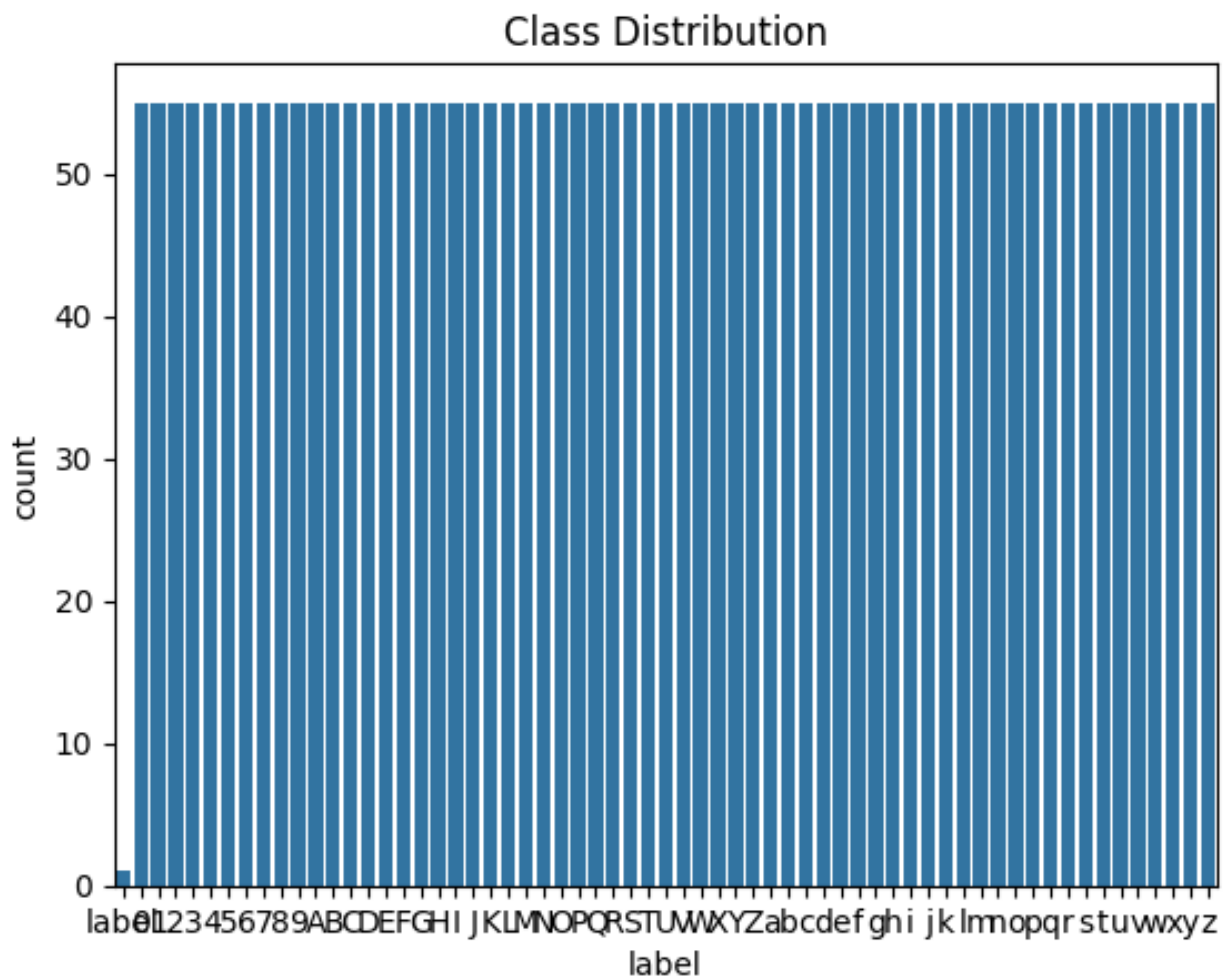


Figure 6: Digit Distribution

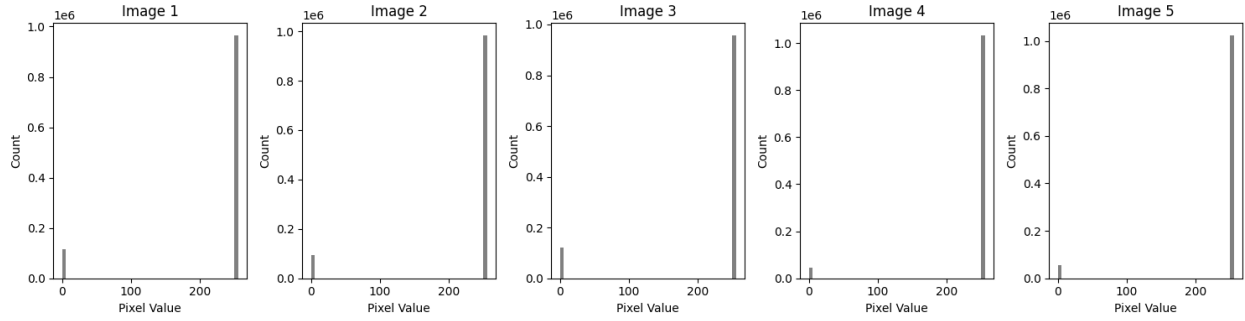


Figure 7: Pixel Intensity Analysis

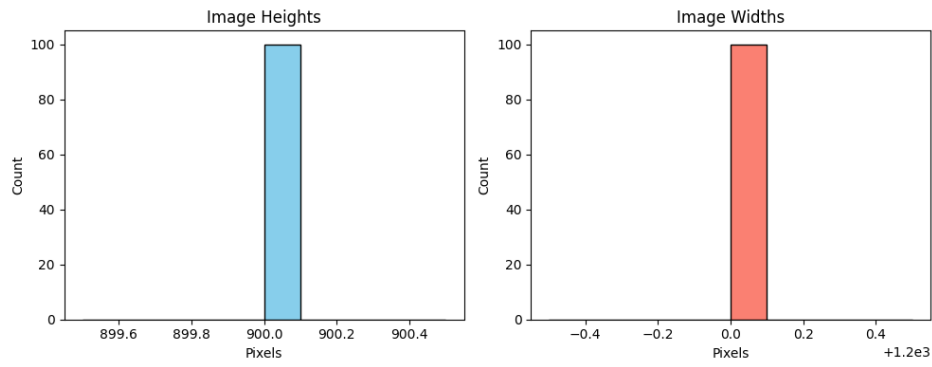


Figure 8: Height Weight Distribution

0.3 Classification of Email Spam

Unnamed: 0	label	text	label_num	
0	605	ham	Subject: enron methanol ; meter # : 988291\nth...	0
1	2349	ham	Subject: hpl nom for january 9 , 2001\n(see a...	0
2	3624	ham	Subject: neon retreat\nho ho ho , we ' re arou...	0
3	4685	spam	Subject: photoshop , windows , office . cheap ...	1
4	2030	ham	Subject: re : indian springs\nthis deal is to ...	0
...	...	...	...	...
5166	1518	ham	Subject: put the 10 on the ft\nthe transport v...	0
5167	404	ham	Subject: 3 / 4 / 2000 and following noms\nhpl ...	0
5168	2933	ham	Subject: calpine daily gas nomination\n>\n>\nj...	0
5169	1409	ham	Subject: industrial worksheets for august 2000...	0
5170	4807	spam	Subject: important online banking alert\ndear ...	1

5171 rows x 4 columns

Figure 9: Dataset Columns

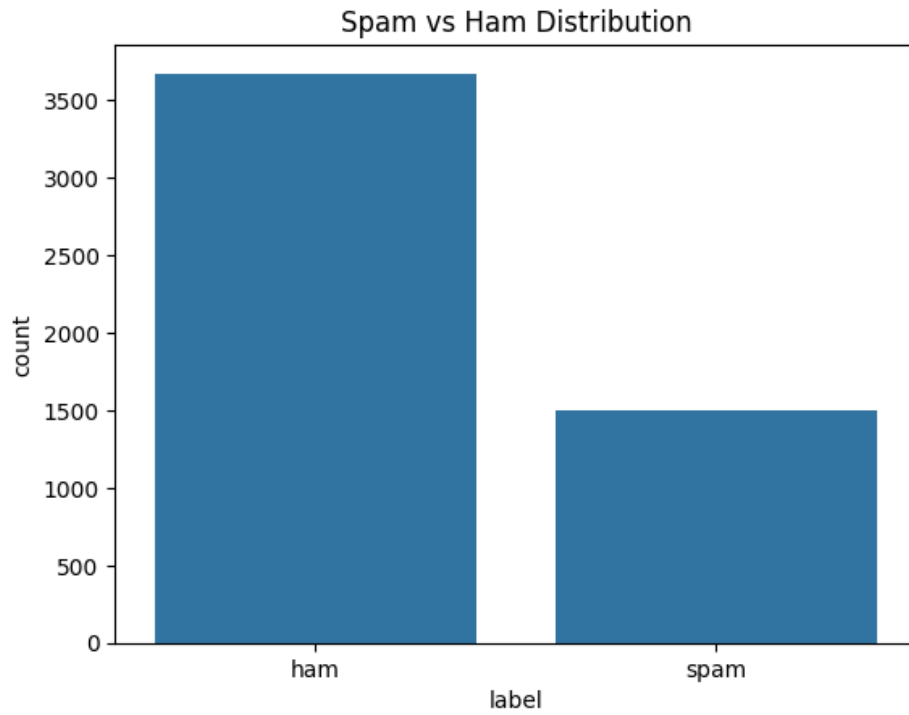


Figure 10: Spam vs Ham Distribution

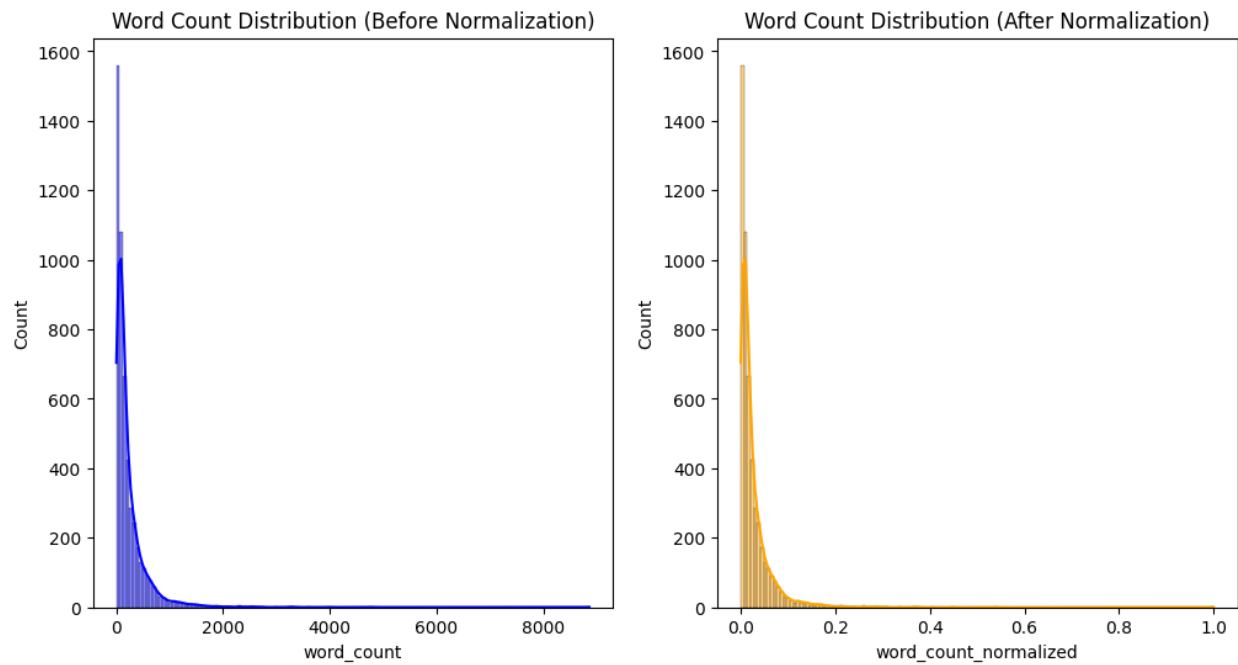


Figure 11: Word Count Distribution (Normalisation)

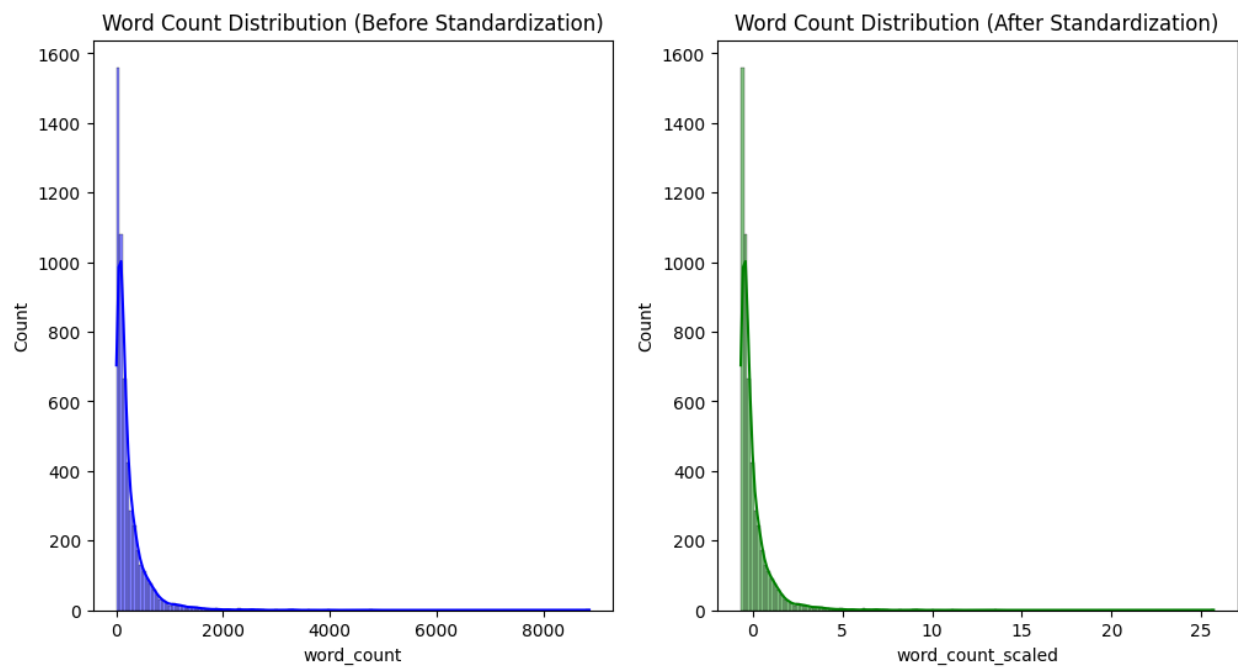


Figure 12: Word Count Distribution (Standardisation)

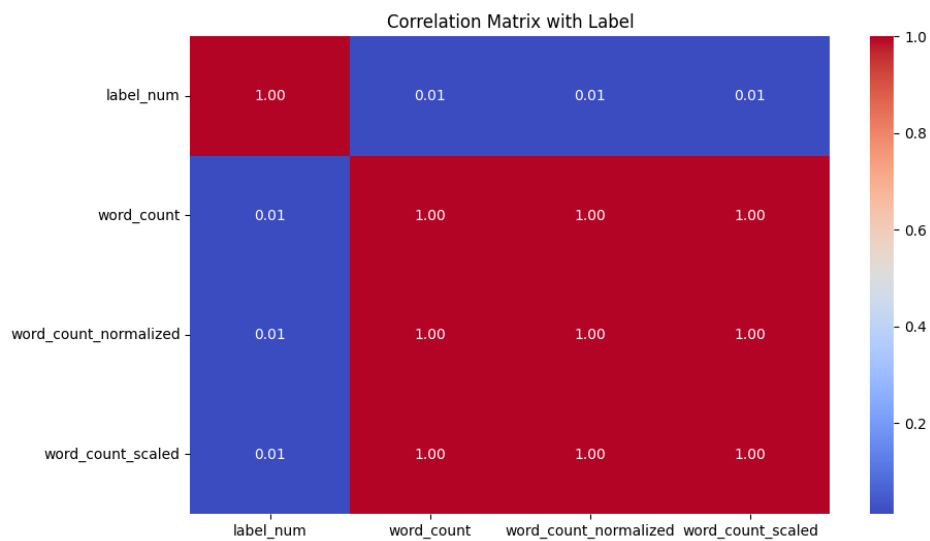


Figure 13: Correlation Matrix

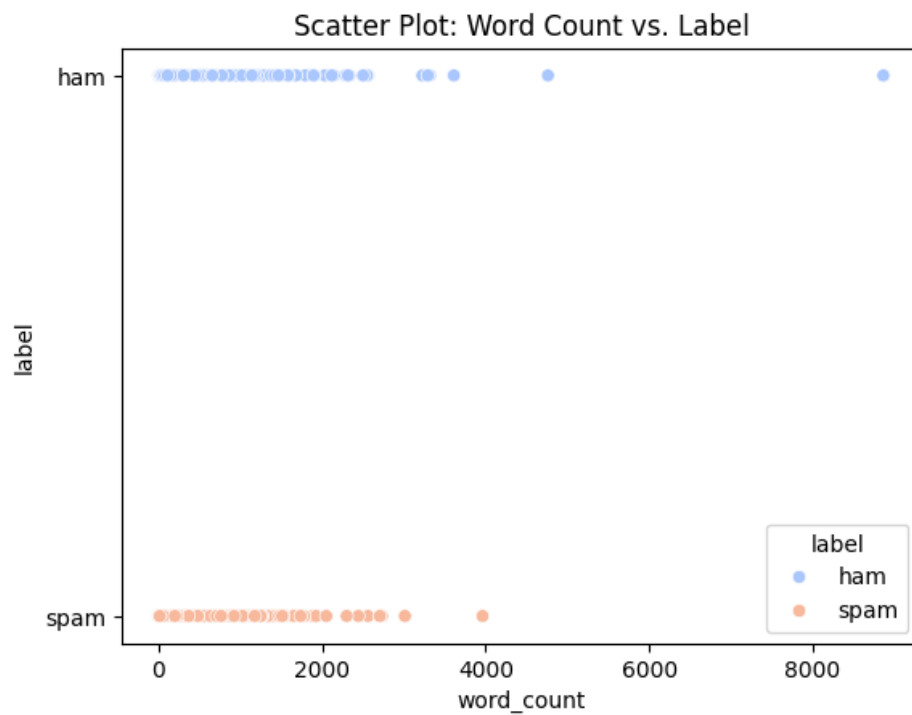


Figure 14: Scatter Plot

0.4 Predicting Diabetes

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
0	Female	80.0	0	1	never	25.19	6.6	140	0
1	Female	54.0	0	0	No Info	27.32	6.6	80	0
2	Male	28.0	0	0	never	27.32	5.7	158	0
3	Female	36.0	0	0	current	23.45	5.0	155	0
4	Male	76.0	1	1	current	20.14	4.8	155	0
...	...	...	...	...	...	...	...	...	...
99995	Female	80.0	0	0	No Info	27.32	6.2	90	0
99996	Female	2.0	0	0	No Info	17.37	6.5	100	0
99997	Male	66.0	0	0	former	27.83	5.7	155	0
99998	Female	24.0	0	0	never	35.42	4.0	100	0
99999	Female	57.0	0	0	current	22.43	6.6	90	0

100000 rows x 9 columns

Figure 15: Dataset Columns

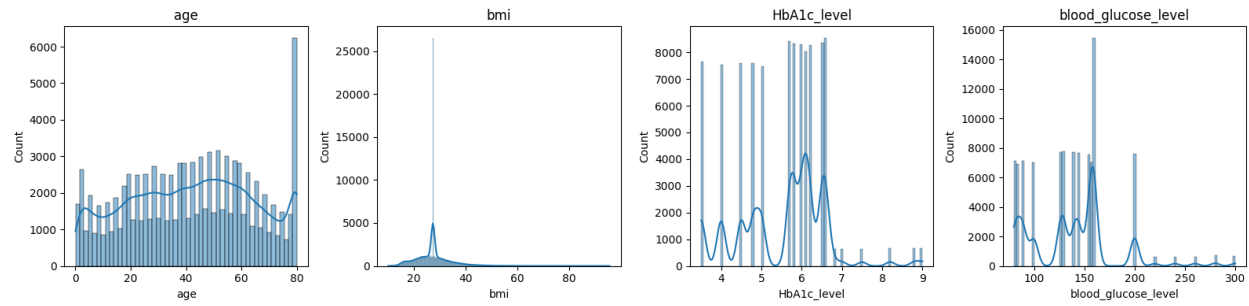


Figure 16: Histogram Distribution

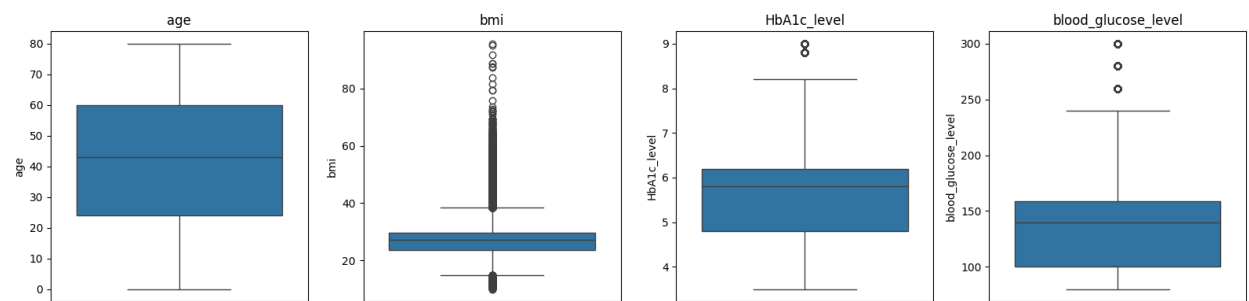


Figure 17: Boxplot distribution

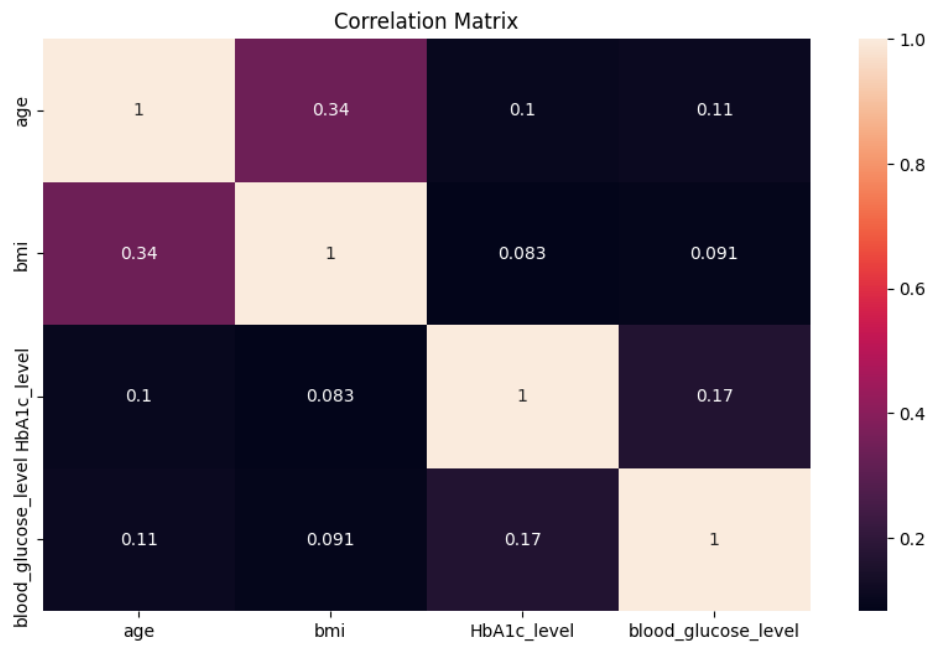


Figure 18: Correlation Matrix

0.5 Iris Dataset

	sepal_length	sepal_width	petal_length	petal_width	species
0	NaN	NaN	NaN	NaN	Species
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Figure 19: Dataset Columns

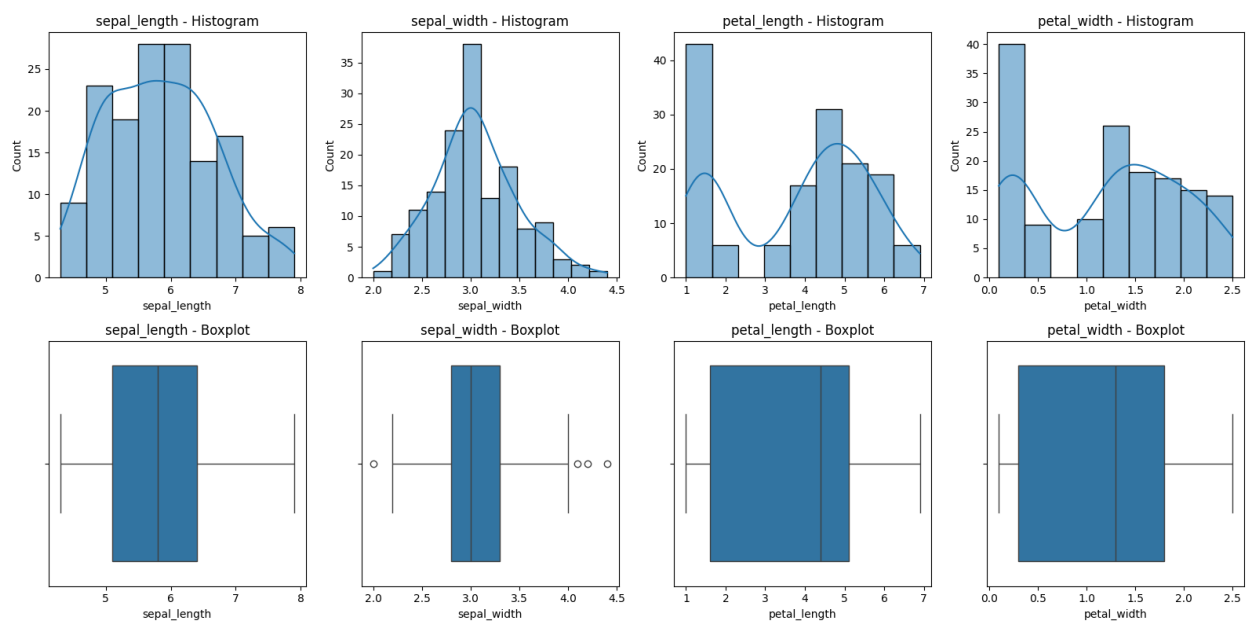


Figure 20: Histogram and Boxplot Distribution

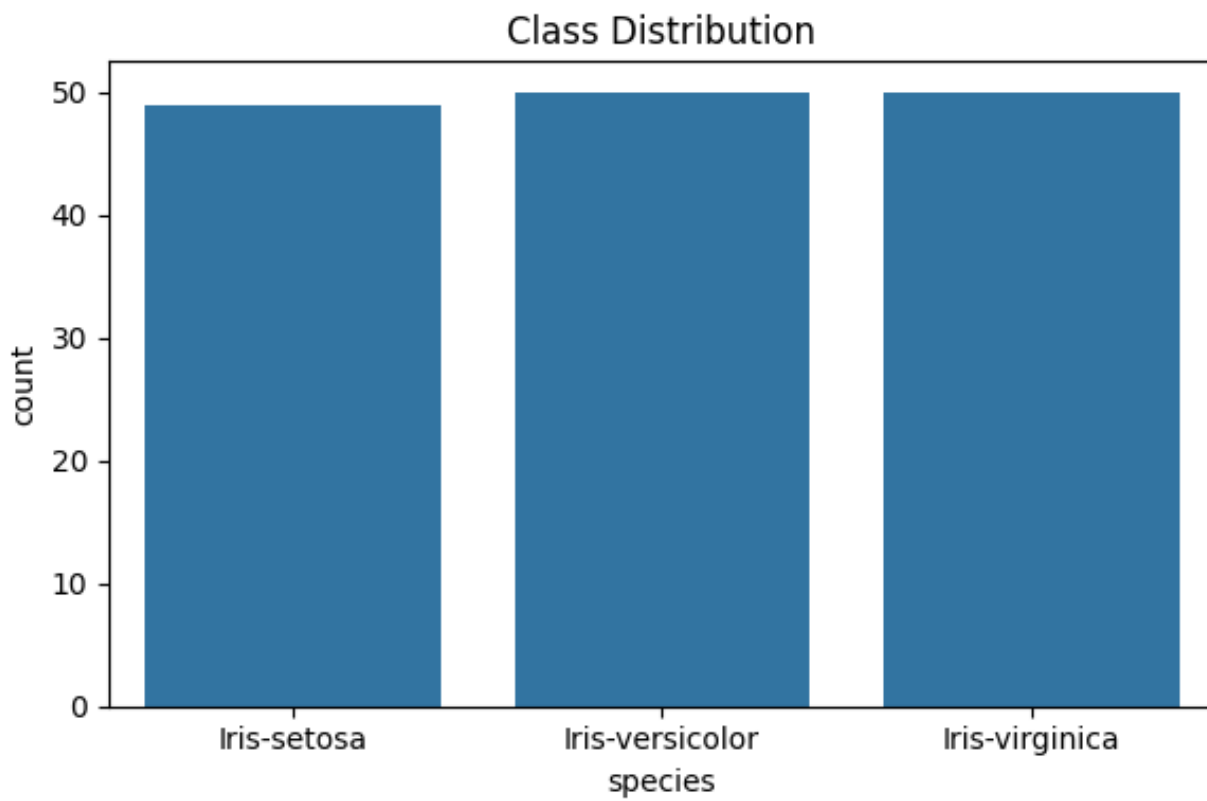


Figure 21: Class Distribution

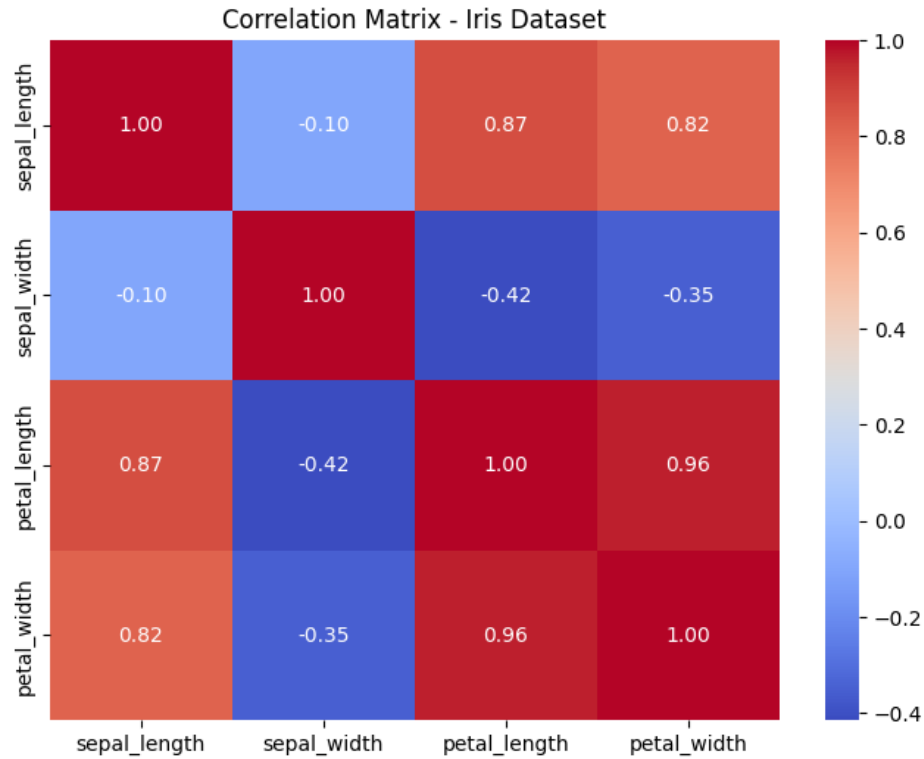


Figure 22: Correlation Matrix

Dataset	Type of ML Task	Feature Selection Technique	Suitable ML Algorithm
Iris Dataset	Multi-class Classification	Correlation Matrix / ANOVA F-value	k-Nearest Neighbors (k-NN), Decision Trees
Loan Amount Prediction	Regression	Recursive Feature Elimination (RFE) / Pearson Correlation	Linear Regression, Random Forest Regressor
Predicting Diabetes	Binary Classification	Chi-Square Test / SelectKBest	Logistic Regression, Support Vector Machine (SVM)
Classification of Email Spam	Binary Classification (NLP)	Information Gain / Chi-Square (on word vectors)	Naive Bayes, SVM
Handwritten Character Recognition / MNIST	Multi-class Image Classification	Principal Component Analysis (PCA)	Convolutional Neural Networks (CNN), SVM

Learning Practices:

- Investigate dataset structure: Comprehend the organization by evaluating dimensions, variable types, and null value occurrences.
- Study data characteristics: Build proficiency in pattern visualization through histograms, boxplots, and correlation heatmaps.
- Evaluate class distribution: Examine label equilibrium and its impact on predictive performance and model reliability.
- Examine variable dependencies: Study interdependencies between attributes using pairwise visualizations and correlation metrics.